

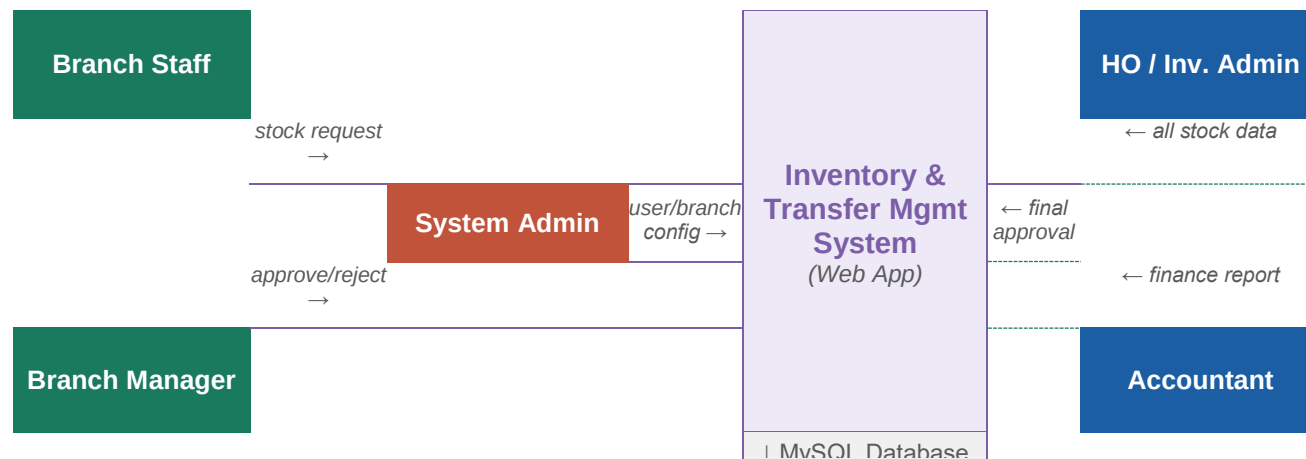
Multi-Branch Inventory & Transfer Management System

Data Flow Diagrams

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1. Context Diagram (Level 0 DFD)

The context diagram shows the system boundary and all external actors. Solid lines indicate input flows into the system; dashed lines indicate output flows from the system to actors.



Flow Legend

	Data input / write flow (actor or process sending data)
	Data output / read flow (system returning data to actor)
Actor box (coloured)	External entity interacting with the system
Process box (purple)	System process (P1–P6)
Store box (gray)	Data store (database table or table group)

2. Level 1 DFD – System Processes

The Level 1 DFD decomposes the system into six core processes. Each row shows an actor, the data it sends in (solid underline), the process that handles it, the data the process returns (dashed underline), and the data stores it reads from or writes to.

Actor	Input flow →	Process	← Output flow	Data Store(s)
All Users	<u>credentials</u>	P1 Authenticate User	JWT token	DS1: Users
HO Admin	<u>catalogue edits</u>	P2 Manage Inventory	stock data	DS3: Items DS4: Stock Lvl's
Staff / Manager	<u>transfer request</u>	P3 Transfer Workflow	status + receipt	DS4: Stock DS5: Transfers
Manager / HO	<u>approve/reject</u>	P4 Approval Engine	pending queue	DS5: Transfers DS6: Audit
Accountant	<u>cost entry</u>	P5 Finance & Costing	finance reports	DS5: Transfers DS7: Costs
System Admin	<u>user/branch data</u>	P6 User & Branch Mgmt	confirmation	DS1: Users DS2: Branches

Data Store Reference

DS1	users	DS2	branches
DS3	items	DS4	stock_levels
DS5	transfer_requests	DS6	audit_logs
DS7	transfer_costs		system_config (global)

Process Descriptions

ID	Process Name	Description
P1	Authenticate User	Validates email/password credentials, issues JWT access token. Enforces RBAC on every subsequent API call. Reads DS1 (users).
P2	Manage Inventory	HO Admin adds/edits items in the master catalogue and records manual stock adjustments. Reads/writes DS3 (items) and DS4 (stock_levels). Flags low-stock items against minimum threshold.
P3	Transfer Workflow	Staff or Manager submits a transfer request. System validates stock availability, calculates total value, determines approval tier, creates the request (PENDING) and progresses it through IN_TRANSIT and RECEIVED to COMPLETED. Reads/writes DS4 and DS5.

P4	Approval Engine	Routes transfer requests to the correct approval tier based on quantity and value thresholds from system_config. Branch Manager performs Level-1 approval; HO Admin performs Level-2. Writes status and comments to DS5; appends every status change to DS6 (audit_log).
P5	Finance & Costing	Accountant views completed/approved transfers and attaches cost records (transport, handling, insurance). Reads DS5; writes DS7 (transfer_costs). Produces finance summary reports filterable by date and branch.
P6	User & Branch Mgmt	System Admin creates, edits, and deactivates user accounts and branch records. Reads/writes DS1 (users) and DS2 (branches). Admin cannot perform approvals.